

Aditya Shibu

GitHub: github.com/adityashibu
Portfolio Website: <https://adityashibu.vercel.app/>

Email: adityashibuonline@gmail.com
Mobile: +971 54 354 7825
LinkedIn: [linkedin.com/in/adityashibu](https://www.linkedin.com/in/adityashibu)
Dubai, UAE

SUMMARY

Final-year CS/AI student with hands-on research experience across the full robotics stack: perception, planning, control, and edge deployment. Built and optimized real-time inference pipelines in C++, CUDA, and TensorRT on NVIDIA Jetson and L40s hardware. Two published conference papers and a journal preprint targeting RA-L. 1st Place at Formula Student AI.

EDUCATION

BSc (Hons) Computer Science with Artificial Intelligence **GPA: 4.0/4.0, First Class Honours**
Heriot Watt University, Dubai, UAE *Sept 2023 - Jul 2026*

RESEARCH AND PUBLICATIONS

- Simulating an Autonomous System in CARLA Using ROS2** **Research Paper** - Published
International Conference on Signal Processing and Information Security (ICSPIS) *November 2025*
- SkySim: Natural Language Control of Drone Swarms using LLMs** **Research Paper** - Published
International Conference on Drones and Unmanned Systems (DAUS) *February 2026*
- LiZIP: A Deep Learning Model for Compression of LiDAR point clouds** **Research Paper** - Preprint
School of Mathematical and Computer Sciences - Heriot Watt University *Target - RA-L*

SKILLS

- Languages:** C, C++, Python, CUDA, JavaScript, Java, OCaml
- AI / Robotics:** PyTorch, TensorFlow, TensorRT, YOLO, ROS2, CARLA, Gazebo, Ollama
- Tools:** Linux, Git, GitHub Actions, Docker

WORK EXPERIENCE

- Center for AI and Robotics (CAIR) - New York University** **Abu Dhabi, UAE**
Research Collaborator (Remote / Part-Time) *Dec 2025 - Feb 2026*
 - Architected and trained a **Conditional Imitation Learning (CILRS)** neural network using PyTorch to navigate complex urban intersections based on high-level GPS commands.
 - Configured and managed headless CARLA simulations on dual **NVIDIA L40s HPC** infrastructure, building synthetic urban driving datasets (RGB + Telemetry) for end-to-end driving models.
- Lab for AI and Robotics Research (LAIRR) - Heriot Watt University** **Dubai, UAE**
Research Intern (Hybrid / Part-Time) *Nov 2025 - Present*
 - Engineered **LiZIP**, a novel auto-regressive neural compression architecture for LiDAR point clouds, writing the full inference pipeline from scratch in C++, custom CUDA kernels, and TensorRT.
 - Optimized for bare-metal edge deployment on the NVIDIA Jetson AGX Orin, achieving real-time inference latency of **16.7ms to 26.8ms** per frame, **outperforming LASzip by up to 2x in speed** while consuming <20% CPU and <1% GPU utilization.
 - Successfully outperformed industry standards **LASzip** and **Google Draco** across both datasets in compression efficiency, reducing payload sizes to **186 KB and 605 KB** respectively, while maintaining a near-lossless maximum error of **<0.017 mm**.
 - Built the perception system for **CogWaters**, a multi-institutional **Dubai RDI-funded** project (Heriot-Watt Dubai, Edinburgh, UK National Robotarium) developing cognitive architectures for autonomous debris-collecting USVs — collected and annotated simulation datasets, trained a **YOLOv8** model, and optimized inference with **TensorRT** for real-time detection.
- AimbeyondD - Heriot Watt University** **Dubai, UAE**
Research Technician (On-Site / Full Time) *Nov 2025 - Present*
 - Led development of **Autonomous Underground Infrastructure Intelligence System (AUIIS)** in collaboration with the **Roads and Transport Authority (RTA) Dubai** for the **Dubai RDI Fund**.
 - Built a multi-agent system combining drone swarms and wheeled robots for underground infrastructure inspection and maintenance.
 - Developed a **ROS 2**-based software architecture for multi-robot coordination and control in **Gazebo Ignition** simulation.
 - Developed **SkySim**, a ROS 2 Jazzy-based simulation environment on Gazebo for Natural Language Control of Drone Swarms using LLMs, with **Artificial Potential Fields**-based collision avoidance and formation flying
 - Deployed on **10x Bitcraze 2.1 Brushless Drones** using **Optitrack** motion capture for real-time position streaming to SLM

- **Atlas Racing FS** Dubai, UAE
 - *Head of Driverless & Lead Autonomous Systems Developer (On-Site / Part-Time)* *May 2025 - Present*
 - Built a perception pipeline using **ZED2i stereo cameras and RoboSense Helios 16 LiDAR**, implementing **DBSCAN clustering** and sensor fusion for object detection and localization
 - Optimized **YOLO inference with TensorRT acceleration**, reducing latency by **88%** (140ms → 16ms) and achieving stable **40-50 FPS** for real-time autonomous vehicle operation
 - Led technical development resulting in **1st Place Overall Statics** and **1st Place Real World AI**, demonstrating production-ready autonomous systems performance
 - Scaled the driverless division from a **5-person founding team to 60+ members across design, outreach, and engineering subsections**, establishing organizational structure and mentoring new recruits.

PROJECTS

- **Kernal Intelligence through Expert Fine-tuning (KITE)**: A fine tuned Small Language Model (SLM) that has been trained on a handmade dataset of Python and CUDA pairs to train the SLM to convert unoptimized Python code to optimized CUDA code | **Tech: Python, OpenAI Triton, CUDA** (May 2025) | **To be released soon**
- **Hydrakon V1 and V2**: Worked with a team of 9 to build an autonomous racing car (Simulation and Hardware Integration) for Formula Student UK | **Tech: CARLA, YOLO, ROS2, Cartographer, Python, C** (April 2025) | <https://github.com/Atlas-Racing>
- **Local RAG**: Developed a fully local RAG with Ollama and LangChain | **Tech: Python, Ollama, and LangChain** (January 2025) | <https://github.com/jonathanjthomas/GDG-RAG-Demo>
- **Space Debris Classification**: Developed a space debris classification model using YOLO, enhancing training data by combining datasets from Roboflow for broader coverage | **Tech: Python, Jupyter Notebook, YOLO** (December 2024) | <https://github.com/adityashibu/spaceDebrisDetection>

AWARDS AND CERTIFICATIONS

- **A Beginner's Guide to Autonomous Robots** - NVIDIA - March, 2026
- **Babcock FS-AI Overall Statics Award Winner** - IMechE - July, 2025
- **FS-AI Real World Award Winner** - IMechE - July, 2025
- **Deep Learning Essentials Intermediate** - IBM - January, 2025
- **GitHub Foundations certification** - GitHub - December, 2024

LEADERSHIP AND VOLUNTEERING EXPERIENCE

- **Google Developer Groups** Dubai, UAE
 - *Lead of AI and ML* *May 2025 - Present*
 - Promoted from Co-lead to Lead after organizing a successful workshop with **120+ attendees**, demonstrating exceptional event management and technical presentation skills
- **Heriot Watt University** Dubai, UAE
 - *MACS Department Officer* *August 2025 - Present*
 - Lead a team of **12 class representatives** across all MACS courses, serving as primary student advocate in faculty committees and implementing policy changes impacting **600+ students**